

2023

Bachelor in Business Administration

Course Title: Financial Derivatives

Code No: **BNK202** Semester: **8th Semester**

Full Marks: 60 Pass Marks: 24 Time: in the next six months. Current price of stock is Rs 300 per share. The put premium is Re 4 per share. Should you exercise the put option and sell the stock if its price decreases to Rs 260? What should be your gain (loss) if you exercise the put option and then immediately purchased the stock? Should you exercise the put option and sell the stock if its price increases to Rs 360? Why? 13. Consider a stock of RK Company that trades for Rs 230 per share. Call options on this stock with the exercise price is Rs 200 per share are currently selling for Rs 45. The calls have an exercise price of Rs 180 and they expire in 3 months. The risk-free rate per annum, and the standard deviation of the stock's return is 0.5. Determine the value of a call option with Black and Scholes- Merton model. Suggest your investment strategy if it is trading at Rs 45. 14. Consider a futures contract in which the current futures price is Rs 414. The initial margin requirement is Rs 20 per contract, and the maintenance margin requirement is Rs 15 per contract. You go long 10 contracts and meet all margin calls but do not withdraw any excess margin. Assume that on the first day, the contract is established at the settlement price, so there is no mark-to-market gain or loss on that day. Calculate the daily gain or loss, cumulative gain or loss, margin balance and margin call if futures prices are 416, 422, 418, 415 and 408 from day 1 to day 5 respectively. Determine the price level that would trigger a margin call. If investor does not deposit margin call amount, what will happen? 15. You simultaneously purchase 100 shares of an underlying stock priced at Rs 77 and write a call option on it with an exercise price of Rs 80 and selling at Rs 6. a. What is the term commonly used for the position that you have taken? b. Determine the profit or loss of your strategy if stock prices at expiration are Rs 70, Rs 75, Rs 80, and Rs 85. c. Determine the maximum profit, maximum loss and break-even price of the strategy. 16. A stock is expected to pay a dividend of Rs 10 per share in 2 months and in 5 months. The stock price is Rs 500, and the risk-free rate of interest is 8% per annum with continuous compounding for all maturities. An investor has just taken a short position in a 6-month forward contract on the stock. a. What are the forward price and the initial value of the forward contract? b. Three months later the price of stock is Rs 480 and risk-free rate of interest is still 8% per annum. What are the forward price and the value of short position in the forward contract? Section "C" Comprehensive answer questions: $[2 \times 10 = 20]$ 17. XYZ firm's stock is currently priced at Rs 200 per share. The option is available on this stock with the exercise price is Rs 200 per share. Six months from now its price will be either Rs 223.6 or Rs 178.88. If the price rises to Rs 223.6, then six months later the price will be either Rs 250 or Rs 200. If, however, the price initially falls to Rs 178.88, then six months later the price will be either Rs 200 or Rs 160. The risk-free rate is 4.08% over the six month period. Using the binomial option-pricing model, what is the fair value of a one-year call option on XYZ's stock? Suppose the actual call is selling for Rs 10, what would be your investment strategy. 18. An asset manager wishes to enter into a two-year equity swap in which he will receive the rate of return on the Stock Index in exchange for paying a fixed interest rate. The Stock Index is at 1150.89 at the beginning of the swap. The swap calls for semiannual payments. a. Calculate the annualized fixed rate on the swap. The current term structure of interest rates is as follows: Term LIBOR $\begin{matrix} (180) & (360) & (540) & (720) \end{matrix}$ 4.58% 5.28% 6.24% 6.65% b. Calculate the market value of the swap 180 days later if the new term structure is Term LIBOR $\begin{matrix} (20) & (200) & (380) & (560) \end{matrix}$ 5.44% 6.29% 6.76% 6.97% The Stock Index is at 1204.10. The notional principal of the swap is Rs 10000000. c. Equity swap is the one of the important type of swaps and it is used to manage the risk of the securities. In this regard, explain what is equity swap.

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Value of derivative securities depend on value of underlying asset.

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2. At the money put option has zero intrinsic value.
 3. According to principle of put option, maximum value of American put is the exercise price.
 4. A riskless hedge involving stock and puts requires a long position in stock and a long position in puts.
 5. A protective put with a higher exercise price has a higher breakeven price.
 6. If the initial margin is Rs 5,000, the maintenance margin is Rs 3,500 and your margin balance is Rs 3,100, in this case you will receive margin call and you must deposit Rs 400 at brokerage firm.
 7. The price of commodity futures contract is the spot price compounded to expiration at the risk-free rate plus the storage costs.
 8. Like interest rate and currency swaps, equity swap payments are always positive.
 9. Forward contract usually has a single delivery date.
 10. The current price of stock is Rs 100, exercise price is Rs 100, risk-free rate 5%, stock pays Re 1 dividend in the next instant, and maturity period of 1 year. The lower bound of the call is Rs 3.76.

Section B

Short Answer Questions:[6 × 5 = 30]

11. What are the reasons of using financial derivatives? Describe the three ways in which derivatives can be misused.
12. Suppose you own a put option that permits you to sell 100 shares of the stock of XYZ Company for Rs 300 per share any time in the next six months. Current price of stock is Rs 300 per share. The put premium is Re 4 per share. Should you exercise the put option and sell the stock if its price decreases to Rs 260? What should be your gain (loss) if you exercise the put option and then immediately purchased the stock? Should you exercise the put option and sell the stock if its price increases to Rs 360? Why?
13. Consider a stock of RK Company that trades for Rs 230 per share. Call options on this stock with the exercise price is Rs 200 per share are currently selling for Rs 45. The calls have an exercise price of Rs 180 and they expire in 3 months. The risk-free rate per annum, and the standard deviation of the stock's return is 0.5. Determine the value of a call option with Black and Scholes- Merton model. Suggest your investment strategy if it is trading at Rs 45.
14. Consider a futures contract in which the current futures price is Rs 414. The initial margin requirement is Rs 20 per contract, and the maintenance margin requirement is Rs 15 per contract. You go long 10 contracts and meet all margin calls but do not withdraw any excess margin. Assume that on the first day, the contract is established at the settlement price, so there is no mark-to-market gain or loss on that day. Calculate the daily gain or loss, cumulative gain or loss, margin balance and margin call if futures prices are 416, 422, 418, 415 and 408 from day 1 to day 5 respectively. Determine the price level that would trigger a margin call. If investor does not deposit margin call amount, what will happen?
15. You simultaneously purchase 100 shares of an underlying stock priced at Rs 77 and write a call option on it with an exercise price of Rs 80 and selling at Rs 6.
 - a. What is the term commonly used for the position that you have taken?
 - b. Determine the profit or loss of your strategy if stock prices at expiration are Rs 70, Rs 75, Rs 80, and Rs 85.
 - c. Determine the maximum profit, maximum loss and break-even price of the strategy.
16. A stock is expected to pay a dividend of Rs 10 per share in 2 months and in 5 months. The stock price is Rs 500, and the risk-free rate of interest is 8% per annum with continuous compounding for all maturities. An investor has just taken a short position in a 6-month forward contract on the stock.
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price and the initial value of the forward contract? b. Three months later the price of stock is Rs 480 and risk-free rate of interest is still 8% per annum. What are the forward price and the value of short position in the forward contract?

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